



LIANGSHAN ENTERPRISE

CONTINGENCY PLANS

Date: 20th – 22nd Nov, 2024

Submitted to: M. Abdullah Abid





LIANGSHAN ENTERPRISE

DISASTER RECOVERY PLAN



Date: 21st Nov, 2024

Submitted to: M. Abdullah Abid

vi. Disaster Recovery Plan (DRP)

1. Introduction

This Disaster Recovery Plan (DRP) provides Liangshan Enterprise with a structured framework to prepare for, respond to, and recover from disasters that may disrupt critical operations. As a global leader in architecture, urban planning, and smart city initiatives, Liangshan Enterprise relies heavily on its IT infrastructure to maintain operations, deliver services, and safeguard sensitive data.

This document outlines the necessary steps to minimize downtime, protect critical assets, and ensure the safety of employees during and after a disaster. It is designed to address various scenarios, including natural disasters, cyberattacks, and system failures, ensuring the organization's resilience in the face of adversity.

2. Definition of a Disaster

At Liangshan Enterprise, a disaster is defined as any unforeseen event, whether caused by natural forces, technological failures, or human actions, that severely impacts the organization's ability to function. Such events may include:

2.1 Disaster Scenarios

- **Loss of Critical Systems:** Malfunction or failure of one or more key IT systems (e.g., servers, network infrastructure, or databases).
- **Facility Disruption:** Extended inaccessibility of office buildings, even if systems remain functional.
- **Combined Impact:** Situations where physical facilities & IT systems are unavailable.
- **Global Events:** Widespread disruptions such as pandemics or geopolitical conflicts.

2.2 Potential Disaster Types

- **Natural Disasters:** Earthquakes, floods, hurricanes, or pandemics.
- **Cybersecurity Incidents:** Ransomware attacks, DDoS attacks, or data breaches.
- **Physical Threats:** Fire, theft, vandalism, terrorism, or civil unrest.
- **Technological Failures:** Power outages, hardware malfunctions, software corruption.

3. Purpose

The primary purpose of this Disaster Recovery Plan is to ensure that Liangshan Enterprise can effectively respond to disasters, minimize disruption, and resume operations as quickly as possible. This plan aims to:

1. Protect the safety and well-being of employees, clients, and stakeholders.
2. Safeguard critical assets, including data, IT infrastructure, and intellectual property.
3. Minimize downtime and financial losses during recovery.
4. Maintain trust and transparency with clients and partners.
5. Establish a clear, actionable framework for disaster recovery activities.

3.1 Objectives

- Prevent the loss of critical organizational resources.
- Ensure data integrity and availability through robust backup and recovery mechanisms.
- Provide a roadmap for coordinating recovery efforts across departments.
- Regularly update and test the DRP to ensure its effectiveness.

4. Scope

The scope of this DRP encompasses all critical IT systems, facilities, and business functions necessary for Liangshan Enterprise to maintain operations. This includes:

- **IT Infrastructure**
 - **Network Infrastructure:** Routers, switches, firewalls, and VPN systems.
 - **Servers and Data Centers:** On-premises and cloud-based servers, including backup data centers.
 - **End-User Devices:** Workstations, laptops, and mobile devices used by employees.
- **Applications and Software**
 - Core business applications such as BIM, CAD, ERP, and project management tools.
 - Collaboration platforms (e.g., Microsoft Teams, Slack).
- **Data Assets**
 - Databases containing client information, architectural designs, and intellectual property.
 - Regular backups stored in offsite and cloud locations.
- **Facilities and Physical Assets**
 - Primary and regional offices in Beijing, London, New York, Dubai, and design studios.
 - Specialized equipment, such as AI-enabled 3D modeling tools and visualization labs.
- **Human Resources**
 - Employees, contractors, and third-party vendors who manage or rely on IT systems.

For disasters outside the scope of this document, reference Liangshan's **Business Continuity Plan (BCP)** or contact the **Business Continuity Lead** at the following:

Name	Title	Location	Phone	Email
Victor Liu	Business Continuity Lead	Beijing HQ	+86-10-7890-5678	vliu@liangshan.com

5. Version Information & Changes

Maintaining an up-to-date Disaster Recovery Plan (DRP) is essential to ensuring that Liangshan Enterprise remains prepared for any disaster. Any modifications to the DRP will be documented here, and the Disaster Recovery Lead is responsible for maintaining and distributing updated versions.

Name of Person Making Change	Role of Person Making Change	Date of Change	Version Number	Notes
Hannah Li	DR Lead	01/15/2022	1.1	Updated contact information for team leads.
Michael Zhang	Facilities Manager	03/01/2022	1.2	Added procedures for alternate site preparation during extended outages.
Sara Morgan	IT Recovery Manager	07/10/2022	1.3	Included new protocols for server virtualization and cloud recovery.
Ethan Lau	Communications Officer	11/05/2022	1.4	Refined external communication templates for stakeholders.
Priya Shah	Data Recovery Specialist	02/20/2023	1.5	Updated data backup and integrity verification procedures.

Table 6.1 Version Update Table

6. Disaster Recovery Team

The Disaster Recovery Team (DRT) is integral to the successful implementation of the DRP. The team includes individuals with specific roles and responsibilities to ensure smooth recovery from a disaster. Each member’s role is clearly defined to avoid duplication of effort and ensure effective communication and execution.

6.1 Disaster Recovery Lead (Mandatory)

The Disaster Recovery Lead serves as the primary decision-maker during a disaster recovery scenario. This individual has full authority to activate the DRP and oversee the recovery process from start to finish.

- **Responsibilities:**
 - Declare a disaster and initiate the DRP.
 - Coordinate the Disaster Recovery Call Tree and ensure effective communication.
 - Monitor progress across all DR teams and ensure alignment with organizational policies.
 - Regularly conduct DRP tests, reviews, and updates.
 - Provide updates to executive leadership and external stakeholders.

Name	Role/Title	Work Phone	Home Phone	Email
Hannah Li	DR Lead	+86-10-1234-5678	+86-123-456-7890	hli@liangshan.com
Daniel Lee	Assistant DR Lead	+86-10-9876-5432	+86-234-567-8901	dlee@liangshan.com
Olivia Carter	Risk Assessment Lead	+44-20-1122-3344	+44-784-555-6789	ocarter@liangshan.com
Emily Tan	Systems Coordinator	+1-212-345-6789	+1-555-678-9101	etan@liangshan.com
Arjun Patel	Technical Advisor	+971-4-3456-7890	+971-555-789-012	apatel@liangshan.com

Table 6.2 Disaster Recovery Team Contact Information

6.2 Disaster Management Team (Elective)

The Disaster Management Team evaluates the disaster's scope and oversees alignment with the DRP. They ensure the recovery process is effectively coordinated across departments and are responsible for post-recovery analysis.

- **Responsibilities:**

- Assess disaster impact and classify its severity.
- Communicate disaster status and instructions to relevant teams.
- Monitor the progress of recovery efforts and provide support.
- Compile detailed reports on disaster impact and recovery measures.

Name	Role/Title	Work Phone	Home Phone	Email
Michael Zhang	Facilities Manager	+86-10-3456-7890	+86-876-543-2100	mzhang@liangshan.com
Priya Shah	Data Recovery Specialist	+44-20-6789-1234	+44-789-112-3456	pshah@liangshan.com
Ethan Lau	Communications Officer	+1-212-567-8901	+1-678-555-1234	elau@liangshan.com

Table 6.3 Disaster Management Team Contact Information

6.3 Facilities Team (Mandatory)

The Facilities Team ensures that Liangshan’s physical facilities are functional and supports operations at standby locations. They manage employee accommodations and physical resources needed for recovery.

- **Responsibilities:**

- Maintain readiness of standby facilities, ensuring accessibility and functionality.
- Coordinate resources for employee relocation and logistics.
- Assess and document physical damage to infrastructure.
- Collaborate with insurance providers to address losses.

Name	Role/Title	Work Phone	Home Phone	Email
Adam Wong	Facilities Coordinator	+86-10-1122-3344	+86-987-654-3210	awong@liangshan.com
Linda Grace	Logistics Manager	+971-4-6789-1234	+971-567-890-345	lgrace@liangshan.com

Table 6.4 Facilities Team Contact Information

7. Other Teams

To ensure the Disaster Recovery Plan (DRP) is executed effectively, Liangshan Enterprise has established specialized teams. Each team has distinct roles and responsibilities to address various aspects of disaster recovery. The following outlines the structure and function of these teams.

7.1 Network Team

- **Role & Responsibilities**

- Identify and assess non-functioning network services at the primary facility during a disaster.
- Prioritize recovery of critical services with minimal business impact.
- Coordinate with third-party providers for timely restoration of network connectivity.
- Activate and configure network services at the secondary facility in case of migration.
- Provide connectivity in the following order of priority: **Disaster Recovery Teams, C-level staff, IT employees, and other employees.**
- Implement tools and systems necessary for network restoration at both standby and primary facilities.
- Prepare a detailed cost report post-disaster for submission to the Disaster Recovery Lead.

Name	Role/Title	Work Phone	Home Phone	Email
Kevin Ma	Network Manager	+86-10-5555-6789	+86-987-123-4560	kma@liangshan.com
Anika Thomas	Network Analyst	+44-20-4456-7890	+44-789-567-8901	athomas@liangshan.com
Victor Chang	Connectivity Specialist	+1-212-345-7890	+1-555-345-6789	vchang@liangshan.com

Table 6.5 Network Team Contact Information

7.2 Server Team

- **Role & Responsibilities**

- Identify and assess non-functioning servers at the primary facility during a disaster.
- Prioritize recovery of servers with the least business impact.
- Maintain updated patches and backups for secondary servers.

- Ensure compliance with organizational server management policies.
- Implement necessary hardware and tools at both standby and primary facilities.
- Compile a detailed cost summary post-disaster for submission to the Disaster Recovery Lead.

Name	Role/Title	Work Phone	Home Phone	Email
John Patel	Server Manager	+971-4-5678-1234	+971-555-678-901	jpatel@liangshan.com
Emily Huang	Server Administrator	+86-10-6789-1234	+86-987-234-5670	ehuang@liangshan.com

Table 6.6 Server Team Contact Information

7.3 Applications Team

- **Role & Responsibilities**

- Identify and assess non-functioning applications at the primary facility during a disaster.
- Prioritize recovery of applications based on their business impact.
- Maintain updated patches for application servers.
- Implement software necessary to ensure application functionality at both standby and primary facilities.
- Prepare a detailed cost report post-disaster for submission to the DRL.

Name	Role/Title	Work Phone	Home Phone	Email
Sophia Wang	Applications Specialist	+44-20-6789-1234	+44-789-678-3450	swang@liangshan.com
David Clarke	Software Engineer	+1-212-678-9012	+1-555-567-8901	dclarke@liangshan.com

Table 6.7 Application Team Contact Information

7.4 Operations Team

- **Role & Responsibilities**

- Provide employees with necessary tools and resources during a disaster.
- Manage the distribution of supplies and equipment.
- Maintain a detailed log of equipment usage during the recovery process.
- Prepare a comprehensive cost report post-disaster for submission to the DRL.

Name	Role/Title	Work Phone	Home Phone	Email
James Lee	Operations Manager	+86-10-5556-7890	+86-987-567-2345	jlee@liangshan.com
Amanda Brown	Logistics Coordinator	+971-4-6789-2345	+971-567-678-901	abrown@liangshan.com

Table 6.8 Operation Team Contact Information

7.5 Senior Management Team

- **Role & Responsibilities**

- Oversee disaster recovery efforts and make critical business decisions.
- Ensure accountability and transparency in the execution of the DRP.
- Provide strategic direction to ensure alignment with organizational goals.

Name	Role/Title	Work Phone	Home Phone	Email
Catherine Liu	COO	+86-10-4567-8901	+86-987-123-4567	cliu@liangshan.com
Oliver White	CFO	+1-212-567-9012	+1-555-345-6789	owhite@liangshan.com

Table 6.9 Senior Management Team Contact Information

7.6 Communication Team

- **Role & Responsibilities**

- Communicate disaster impact and updates to employees, partners, clients, vendors, and the media.
- Prepare detailed communication reports post-disaster for submission to the Disaster Recovery Lead.

Name	Role/Title	Work Phone	Home Phone	Email
Daniel Grant	Communications Manager	+44-20-2345-6789	+44-789-345-6789	dgrant@liangshan.com
Fiona Yang	Public Relations Officer	+1-212-456-7890	+1-555-234-5678	fyang@liangshan.com

Table 6.10 Communication Team Contact Information

7.7 Finance Team

- **Role & Responsibilities**

- Ensure availability of funds for both disaster-related and routine operational expenses.
- Manage small-scale expenses such as accommodations and incremental bills.
- Oversee large-scale expenses such as facility repairs and new equipment purchases.
- Review and approve spending by disaster teams.
- Facilitate regular payroll processes.
- Coordinate with creditors and banking partners as needed.

Name	Role/Title	Work Phone	Home Phone	Email
Liam Carter	Finance Manager	+86-10-6789-4567	+86-987-234-7890	lcarter@liangshan.com
Zoe Johnson	Financial Analyst	+971-4-5678-9012	+971-567-890-123	zjohnson@liangshan.com

Table 6.11 Finance Team Contact Information

8. Disaster Recovery Call Tree

The Disaster Recovery Call Tree is a structured communication plan ensuring rapid dissemination of critical information during a disaster. It involves a cascading system where individuals at each level of responsibility notify their respective teams, enabling a coordinated response.

Name	Role/Title	Work Phone	Home Phone	Email
Hannah Li	DR Lead	+86-10-1234-5678	+86-123-456-7890	+86-789-012-3456
Daniel Lee	Assistant DR Lead	+86-10-9876-5432	+86-234-567-8901	+86-876-543-2100
Olivia Carter	Risk Assessment Lead	+44-20-1122-3344	+44-784-555-6789	+44-789-345-6789
Emily Tan	Systems Coordinator	+1-212-345-6789	+1-555-678-9101	+1-678-123-4567
Arjun Patel	Technical Advisor	+971-4-3456-7890	+971-555-789-012	+971-567-345-6789

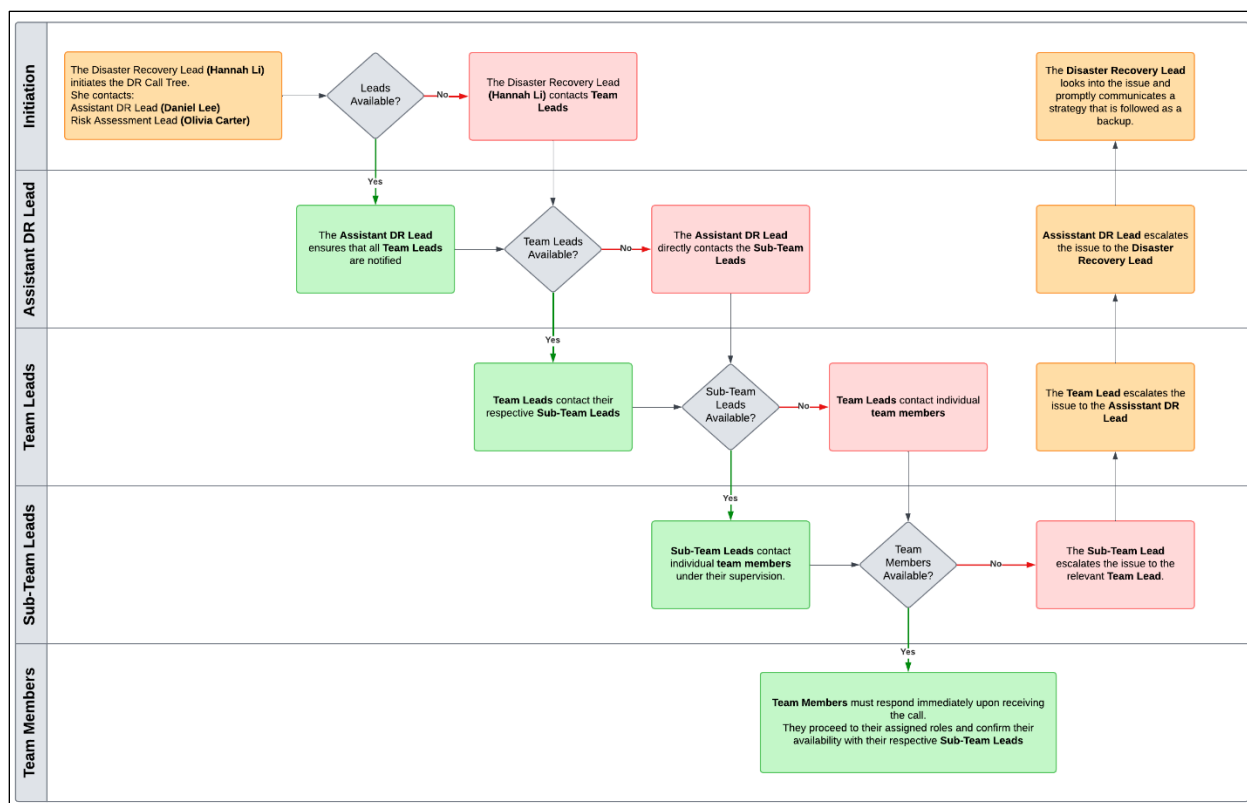
Michael Zhang	Facilities Manager	+86-10-3456-7890	+86-876-543-2100	+86-678-123-4567
Priya Shah	Data Recovery Specialist	+44-20-6789-1234	+44-789-112-3456	+44-123-567-7890
Ethan Lau	Communications Officer	+1-212-567-8901	+1-678-555-1234	+1-789-456-1230
Adam Wong	Facilities Coordinator	+86-10-1122-3344	+86-987-654-3210	+86-234-567-8901
Linda Grace	Logistics Manager	+971-4-6789-1234	+971-567-890-345	+971-123-678-9012
Kevin Ma	Network Manager	+86-10-5555-6789	+86-987-123-4560	+86-345-678-9012
Anika Thomas	Network Analyst	+44-20-4456-7890	+44-789-567-8901	+44-456-789-0123
Victor Chang	Connectivity Specialist	+1-212-345-7890	+1-555-345-6789	+1-678-901-2345
John Patel	Server Manager	+971-4-5678-1234	+971-555-678-901	+971-789-012-3456
Emily Huang	Server Administrator	+86-10-6789-1234	+86-987-234-5670	+86-901-234-5678
Sophia Wang	Applications Specialist	+44-20-6789-1234	+44-789-678-3450	+44-345-678-9012
David Clarke	Software Engineer	+1-212-678-9012	+1-555-567-8901	+1-456-789-0123
James Lee	Operations Manager	+86-10-5556-7890	+86-987-567-2345	+86-567-890-1234
Amanda Brown	Logistics Coordinator	+971-4-6789-2345	+971-567-678-901	+971-789-012-3456
Catherine Liu	COO	+86-10-4567-8901	+86-987-123-4567	+86-234-678-9012
Oliver White	CFO	+1-212-567-9012	+1-555-345-6789	+1-345-678-9012
Daniel Grant	Communications Manager	+44-20-2345-6789	+44-789-345-6789	+44-123-567-8901

Fiona Yang	Public Relations Officer	+1-212-456-7890	+1-555-234-5678	+1-789-345-6780
Liam Carter	Finance Manager	+86-10-6789-4567	+86-987-234-7890	+86-567-890-3456
Zoe Johnson	Financial Analyst	+971-4-5678-9012	+971-567-890-123	+971-234-567-8901

Table 6.12 Disaster Recovery Call Tree Information

9. Disaster Recovery Call Tree Process

The Disaster Recovery Call Tree Process outlines the step-by-step communication sequence to ensure efficient coordination during a disaster. Below is the Liangshan-specific call tree:



10. Communicating During a Disaster

Effective communication is a cornerstone of disaster recovery at Liangshan Enterprise. The following subsections outline the structured processes for communicating with authorities, employees, clients, vendors, and the media during a disaster to ensure transparency, efficiency, and prompt resolution.

10.1 Communicating with the Authorities

The Communications Team is responsible for promptly notifying authorities about the disaster. The notification will include critical details about the event, its impact, and required assistance.

- **Details to Share with Authorities:**
 - **Location:** Liangshan Enterprise Headquarters, Beijing, China.
 - **Nature of Disaster:** Data center flooding due to heavy rainfall.
 - **Magnitude:** Severe impact on critical infrastructure, including server damage.
 - **Impact:** Service disruption affecting multiple clients and internal operations.
 - **Assistance Required:** Emergency drainage services and building safety inspections.
 - **Anticipated Timelines:** Immediate assistance needed within two hours.
- **Authorities Contacts:**

Authority	Point of Contact	Phone Number	Email
Police Department	Inspector Liu Wei	+86-10-2222-1111	lwei@beijingpolice.cn
Fire Department	Captain Zhang Yong	+86-10-3333-4444	zhangy@firedept.cn
Municipal Emergency Services	Manager Chen Lin	+86-10-5555-6666	chenlin@emergencyservices.cn

Table 6.13 Authorities Contact Information

10.2 Communicating with Employees

The Communications Team will notify employees of the disaster and provide them with necessary instructions using prioritized communication channels:

- **Communication Channels (Priority Order):**

1. Corporate email.
2. Personal email.
3. Mobile phone calls.
4. Home phone calls.

- **Employees to Notify:**

Name	Role/Title	Home Phone	Mobile Phone	Email
Sarah Lin	HR Lead	+86-10-7777-8888	+86-123-456-7890	sarah.lin@liangshan.com
Jason Carter	Sr. Software Engineer	+1-212-234-5678	+1-555-678-1234	jason.carter@liangshan.com
Mei Wong	Operations Supervisor	+86-10-5678-3456	+86-987-654-3210	mei.wong@liangshan.com
Zoe Johnson	Finance Analyst	+44-20-6789-2345	+44-789-567-1234	liam.brown@liangshan.com
Priya Desai	Technical Writer	+971-4-5678-9012	+971-567-890-3456	priya.desai@liangshan.com

Table 6.14 Employees to Notify Information

- **Information Provided to Employees:**

- Safety instructions for evacuation or working remotely.
- Alternative work arrangements, including backup office locations or remote access guidance.
- Support services such as counseling or IT assistance.

10.3 Communicating with Clients

The Communications Team will notify crucial clients about the disaster's impact on services. Clients will be contacted via email, followed by phone calls for confirmation.

- **Crucial Clients:**

Company Name	Point of Contact	Phone Number	Email
SinoTech Industries	Mr. Chen Guang	+86-10-1234-9876	chen.guang@sinotech.com
Global FinTech Ltd.	Ms. Laura Davies	+1-212-555-6789	laura.davies@globalfintech.com
AsiaBuild Solutions	Mr. Ramesh Iyer	+971-4-2345-6789	ramesh.iyer@asiabuild.com
EuroData Systems	Ms. Sophie Laurent	+44-20-5678-9012	sophie.laurent@eurodata.com
Pacific Logistics	Mr. Akio Tanaka	+81-3-6789-1234	akio.tanaka@pacificlogistics.jp

Table 6.15 Crucial Clients Contact Information

- **Information Provided to Clients:**
 - Current status of operations and services.
 - Expected timelines for resolution and service restoration.
 - Points of contact for updates.

10.4 Communicating with Vendors

Vendors will be notified to ensure continuity of supplies and services necessary for disaster recovery.

10.4.1 Crucial Vendors:

Company Name	Point of Contact	Phone Number	Email
Alpha Networking Ltd.	Mr. Kevin Chen	+86-10-5678-2345	kevin.chen@alphanet.cn
RapidTech Solutions	Ms. Karen Miles	+1-212-678-9012	karen.miles@rapidtech.com
SecureIT Supplies	Mr. Carlos Rivera	+44-20-7890-1234	carlos.rivera@secureit.com
Zenith Power Backup	Ms. Ayesha Khan	+971-4-6789-4567	ayesha.khan@zenithpower.com

Table 6.16 Crucial Primary Vendors Contact Information

10.4.2 Secondary Vendors:

Company Name	Point of Contact	Phone Number	Email
DataMax Solutions	Mr. Alex Yu	+86-10-7890-1234	alex.yu@datamax.cn
TechCare Enterprises	Ms. Monica Zhang	+1-212-345-6789	monica.zhang@techcare.com
SmartBuild Logistics	Mr. Faizan Ahmed	+971-4-5678-2345	faizan.ahmed@smartbuild.com
EcoConnect Systems	Ms. Emily Harper	+44-20-2345-6789	emily.harper@ecoconnect.com

Table 6.17 Secondary Vendors Contact Information

10.5 Communicating with the Media

The Communications Team will coordinate with the media to release accurate and timely information regarding the disaster. Official statements will ensure transparency while protecting Liangshan Enterprise's reputation.

- **Media Contacts:**

Company Name	Point of Contact	Phone Number	Email
Beijing Daily	Ms. Hui Lin	+86-10-2345-6789	hui.lin@beijingdaily.cn
TechGlobal News	Mr. James Peterson	+1-212-345-6789	james.peterson@techglobal.com
Finance Today	Ms. Rachel Morgan	+44-20-5678-2345	rachel.morgan@financetoday.com

Table 6.18 Media Contact Information

- **Key Responsibilities:**

- Prepare and disseminate press releases.
- Organize press briefings or interviews if necessary.
- Monitor media coverage and address inaccuracies promptly.

11. Recovery Facilities

Liangshan Enterprise has established detailed plans for recovery facilities to ensure continuity of operations during and after disasters. These facilities and processes provide redundancy, mobility, and strategic recovery mechanisms tailored to Liangshan's infrastructure and business needs.

11.1 Mobile Site Recovery Plan

In the event that Liangshan's primary facility becomes inaccessible, a mobile site recovery plan ensures minimal disruption to critical business operations.

- **Notification Protocol:**
 - **Action:** Immediately notify the Disaster Recovery Lead, **Hannah Li** (hli@liangshan.com), detailing the nature, location, and magnitude of the disaster.
 - **Role Description:** Hannah Li is responsible for coordinating the mobile site recovery efforts and reporting progress to senior management.
- **Mobile Site Initiation:**
 - **Action:** Notify the Mobile Site Manager, **Arjun Desai** (adesai@liangshan.com), within the first 24 hours of the disaster.
 - **Report:** Submit a detailed report summarizing operational requirements for the mobile site.
- **Backup Verification:**
 - **Action:** Validate the integrity of all necessary backups using the Backup Validation Protocol.
 - **Objective:** Confirm the readiness of all critical data to be deployed at the mobile site.
- **Equipment Procurement:**
 - **Action:** Initiate procurement of any required hardware or software to ensure operational continuity.
 - **Responsibility:** Procurement Officer **Linda Grace** oversees sourcing and delivery.
- **Logistics and Communication:**
 - **Coordination:** Partner with Logistics Manager **Amanda Brown** to deploy resources to the mobile site.
 - **Communication Infrastructure:** Notify the telecom provider of necessary changes to support the mobile site.
- **Infrastructure Establishment:**
 - **Action:** Set up essential infrastructure, including power and communication systems, at the mobile site location.
 - **Verification:** Confirm readiness for activation upon trailer deployment.

- **Network Rerouting:**
 - **Action:** Redirect network traffic from the primary facility to the mobile site.
 - **Responsibility:** **Victor Chang**, Connectivity Specialist, oversees this process.
- **System Activation:**
 - **Action:** Activate systems and conduct rigorous testing to ensure operational readiness.
 - **Data Loading:** Load verified backups and validate the restoration of critical applications.
- **Operational Resumption:**
 - **Tasks:** Resume normal business operations, adhering to weekly and monthly backup schedules.
 - **Documentation:** Maintain a log of all mobile site activities and conduct periodic evaluations for improvement.

11.2 Hot Site Recovery Plan

Liangshan Enterprise maintains contracts with external hot site providers to ensure rapid recovery of IT operations at a fully equipped secondary location.

- **Disaster Recovery Service Notification:**
 - **Action:** Notify the hot site provider within the first hour of disaster declaration.
 - **Responsible Party:** Disaster Recovery Lead **Hannah Li** initiates this process.
- **Modem Shipment Coordination:**
 - **Action:** Expedite shipment of essential communication equipment to the hot site.
- **Written Confirmation:**
 - **Action:** Submit formal disaster details and recovery activation requests to the hot site provider within 48 hours.
- **Travel Arrangements:**
 - **Action:** Arrange for the Disaster Recovery Team to travel to the hot site.
 - **Coordinator:** Travel arrangements are managed by **Ethan Lau**, Communications Officer.
- **Tapes Availability:**
 - **Verification:** Confirm that all necessary backup tapes are packed and ready for transport to the hot site.
- **Purchase Order Preparation:**
 - **Action:** Process financial requisitions for hot site usage through **Liam Carter**, Finance Manager.

- **Materials Checklist Review:**
 - **Action:** Review the recovery materials checklist to ensure all critical items are included.
- **Information Sharing:**
 - **Action:** Provide recovery details and requirements to the hot site team for seamless operations.
- **Travel Expenses:**
 - **Action:** Ensure provision of necessary travel funds for team members traveling to the hot site.

11.3 Site Rebuilding Process

Following a disaster, Liangshan will implement a structured rebuilding plan for its primary data center. This process emphasizes resilience, security, and operational excellence.

- **Floor Plan Utilization:**
 - **Action:** Use existing blueprints to guide reconstruction.
 - **Enhancements:** Integrate advanced layouts to accommodate emerging technology and business requirements.
- **Hardware Needs Assessment:**
 - **Action:** Evaluate hardware roles and functions to determine replacement or upgrades.
 - **Coordinator:** **Emily Huang**, Server Administrator, leads the hardware assessment.
- **Infrastructure Specifications:**
 - **Data Center Size:** Rebuild to a size of 80,000–100,000 square feet, scalable for future needs.
 - **Power Requirements:** Secure power capacities of 1–5 megawatts.
- **Security Features:**
 - **Locked Area:** Establish a secure area with access controls.
 - **Detection Systems:** Deploy water, smoke, temperature, and motion sensors.
- **Structural Integrity:**
 - **Floor-to-Ceiling Studding:** Implement reinforced walls for enhanced safety.
 - **Raised Flooring:** Install raised flooring for efficient airflow and cable management.
- **Advanced Monitoring and Redundancy:**
 - **Action:** Incorporate redundant power supplies, backup generators, and advanced surveillance systems to minimize future risks.

12. Description of Recovery Facilities

Liangshan Enterprise has established a **Disaster Command and Control Center (DCCC)** located at **88 Industrial Road, Beijing**, approximately 15 miles from the primary facility in downtown Beijing. This facility will serve as the operational hub during a disaster, ensuring uninterrupted decision-making, communication, and IT recovery operations. The standby facility is fully equipped to accommodate the needs of IT and Disaster Recovery (DR) teams while maintaining business continuity.

12.1 Facility Resources

The standby facility provides the following essential resources to ensure operational efficiency during a disaster:

- **Document Accessibility:** Copies of the DRP are maintained for easy reference.
- **Redundant Server Room:** Equipped with state-of-the-art servers and storage infrastructure to support enterprise-level operations.
- **Office Space:** Dedicated office areas for IT and DR teams, enabling seamless operations.
- **Connectivity:** High-speed external data and voice connectivity for uninterrupted communication.
- **Employee Accommodations:** Sleeping quarters for employees working extended shifts.
- **Kitchen Facilities:** Fully stocked kitchen with food supplies, appliances, and utensils.
- **Bathroom Facilities:** Fully functional bathrooms with showers, toilets, and sinks.
- **Parking:** Sufficient parking for employee vehicles, ensuring easy access.

12.2 Transportation to the Standby Facility

To facilitate transportation, arrangements have been made with local taxi providers, rental car companies, and travel agents in Beijing.

- **Taxi Providers:**

Company Name	Address	Phone Number
BeijingRide Taxis	45 Tiananmen Road, Beijing	+86-10-4567-1234
FastCab Beijing	12 Chaoyang Avenue, Beijing	+86-10-6789-3456
QuickTaxi Services	88 Haidian Road, Beijing	+86-10-2345-6789

Table 6.19 Taxi Providers Contact Information

- **Rental Car Providers:**

Company Name	Address	Phone Number
DriveSmart Rentals	19 Wangfujing Street, Beijing	+86-10-1234-5678
Beijing Auto Hire	56 Xicheng Road, Beijing	+86-10-7890-1234
Speedy Car Rentals	34 Dongcheng Avenue, Beijing	+86-10-5678-2345

Table 6.20 Rental Car Contact Information

- **Travel Agents:**

Company Name	Address	Phone Number
Capital Travels	22 Finance Street, Beijing	+86-10-4567-8901
Horizon Travel Agency	9 Central Avenue, Beijing	+86-10-2345-6789
Skyline Travel Services	77 Olympic Road, Beijing	+86-10-6789-3456
Beijing Global Journeys	33 Culture Street, Beijing	+86-10-7890-1234
Sunrise Travel Solutions	55 Garden Avenue, Beijing	+86-10-3456-7890

Table 6.21 Travel Agents Contact Information

12.3 Operational Considerations

The **Facilities Team**, led by **Michael Zhang**, will coordinate accommodations, catering, and standby facility maintenance to ensure optimal working conditions.

- **Accommodations**

Hotel arrangements will be provided for employees requiring extended stays.

Hotel Name	Address	Phone Number
Grand Beijing Hotel	88 Tiananmen Road, Beijing	+86-10-5678-1234
Capital Suites Beijing	12 Haidian Avenue, Beijing	+86-10-2345-7890
Skyline Residency	34 Wangfujing Street, Beijing	+86-10-3456-6789
Horizon Inn	78 Chaoyang Avenue, Beijing	+86-10-4567-1234
Royal Beijing Hotel	56 Xicheng Avenue, Beijing	+86-10-6789-3456
Pearl Continental Beijing	99 Central Street, Beijing	+86-10-7890-2345

Table 6.22 Emergency Hotels Contact Information

- **Catering**

Hotel Name	Address	Phone Number
Beijing Feast Catering	45 Tiananmen Road, Beijing	+86-10-2345-6789
Elite Beijing Caterers	22 Central Avenue, Beijing	+86-10-5678-1234
Gourmet Beijing Services	34 Haidian Road, Beijing	+86-10-7890-3456
Capital Banquets Beijing	88 Olympic Road, Beijing	+86-10-3456-7890

Table 6.23 Emergency Catering Contact Information

- **Standby Facility Maintenance**

Hotel Name	Address	Phone Number
FacilityPro Beijing	67 Central Avenue, Beijing	+86-10-1234-6789
UrbanFix Beijing Services	11 Wangfujing Road, Beijing	+86-10-7890-3456
CarePoint Maintenance	99 Dongcheng Avenue, Beijing	+86-10-5678-2345

Table 6.24 Standby Maintenance Facility Contact Information

12.4 Data and Backups

Liangshan Enterprise ensures data redundancy and security by maintaining regular backups across multiple locations.

Data Name or Group	Data Type	Backup Frequency	Backup Location
Employee Records	HR Data	Daily	Cloud Storage (Aliyun)
Financial Transactions	Finance Data	Hourly	Beijing Data Center A
Customer Contracts	Legal/Contractual	Weekly	Beijing Data Center B
Project Blueprints	Architectural Designs	Weekly	Encrypted External Drives
Security Logs	System Logs	Hourly	SIEM Backup System
ERP Databases	Business Applications	Daily	Remote Hot Site
CRM Databases	Customer Relations Data	Daily	Cloud-Based CRM Backup
Marketing Plans	Strategy Documents	Weekly	Secure Internal Drives
Vendor Agreements	Legal/Contracts	Monthly	Beijing Data Center C
R&D Data	Proprietary Data	Weekly	Encrypted Cloud Repository
Network Configurations	IT Infrastructure	Daily	Remote Hot Site
Application Logs	System Logs	Hourly	SIEM Backup System
Training Manuals	Internal Documentation	Monthly	In-House Backup Server
Disaster Recovery Plans	Documentation	Quarterly	Secure Print and Cloud Copies
Compliance Documents	Audit Records	Weekly	Beijing Data Center D

Table 6.25 Backup Data Information

13. Case 01: Fire Outbreak at Liangshan Enterprise Primary Data Center

13.1 Background

On March 15, 2024, at 2:30 AM, a fire outbreak was detected in Liangshan Enterprise's primary data center located in downtown Beijing. The fire caused extensive damage to critical IT infrastructure, rendering servers, networking equipment, and storage systems non-functional. The fire suppression system activated promptly, but the smoke and heat caused significant disruption to operations. Fortunately, all personnel were safely evacuated, and no injuries were reported. This incident necessitated the activation of Liangshan's Disaster Recovery Plan (DRP).

13.2 Incident Timeline and DRP Activation

- 2:30 AM: Fire Detection

- **Event:** Smoke alarms in the data center triggered the fire suppression system.
- **Immediate Action:**
 - Security personnel alerted the **Facilities Team** led by **Michael Zhang**.
 - Emergency services, including the local fire department, were contacted to extinguish the fire.

- 3:00 AM: DRP Activation

- **Decision:** The **Disaster Recovery Lead (Hannah Li)** declared a disaster and activated the DRP after assessing the extent of the damage.
- **Notification Protocol:**
 - The **Disaster Recovery Call Tree** was initiated.
 - All DR team leads were contacted and instructed to report to the Disaster Command and Control Center (DCCC) at **88 Industrial Road, Beijing**.

- 4:00 AM: Transition to the Standby Facility

- **Action:**
 - The **Facilities Team**, led by **Adam Wong**, prepared the standby facility for operations.
 - Transportation arrangements were made with **BeijingRide Taxis** to ensure key personnel reached the facility.

13.3 Mapping the DRP to the Incident

13.3.1 Description of Recovery Facilities

- **Use of Standby Facility:**
 - The Disaster Command and Control Center (DCCC) at **88 Industrial Road, Beijing**, was activated.
 - Redundant server rooms in the facility ensured continuity of IT operations.
 - Temporary sleeping & kitchen facilities were utilized by IT staff and DR teams.
- **Transportation:**
 - Employees were transported to the DCCC using **BeijingRide Taxis** and **DriveSmart Rentals**.
- **Accommodation and Catering:**
 - **Grand Beijing Hotel** provided accommodations for employees requiring extended stays.
 - **Beijing Feast Catering** supplied meals for on-site teams.

13.4 Data and Backups

- **Action Taken:**
 - Backup data from **Cloud Storage (Aliyun)** and **Remote Hot Site** were verified by the **Server Team**, led by **John Patel**.
 - Encrypted external drives containing critical project blueprints were retrieved and loaded onto the standby facility's servers.
- **Outcome:**
 - Operational data, including CRM databases, ERP systems, and financial records, were restored within six hours.

13.5 Key Teams and Their Actions

13.5.1 Facilities Team

- **Action Taken:**
 - Assessed the structural damage at the primary facility.
 - Coordinated with **UrbanFix Beijing Services** for initial repair estimates.
- **Outcome:**
 - Standby facility infrastructure was fully operational within 2 hours of DRP activation.

13.5.2 Network Team

- **Action Taken:**

- Kevin Ma, the **Network Manager**, rerouted network traffic to the standby facility.
- Anika Thomas, the **Network Analyst**, ensured secure connectivity for critical systems.
- **Outcome:**
 - Network connectivity was restored within 4 hours, prioritizing DR teams, executive staff, and IT personnel.

13.5.3 Server Team

- **Action Taken:**
 - Emily Huang, the **Server Administrator**, restored server functionality using verified backups.
- **Outcome:**
 - All mission-critical servers were operational at the standby facility within 6 hours.

13.5.4 Applications Team

- **Action Taken:**
 - Sophia Wang, the **Applications Specialist**, prioritized recovery of enterprise applications.
 - David Clarke, the **Software Engineer**, deployed updated patches.
- **Outcome:**
 - Core business applications were restored within 8 hours.

13.5.5 Communications Team

- **Action Taken:**
 - Ethan Lau, the **Communications Officer**, informed employees of safety measures and work-from-home protocols via corporate email and phone.
 - Daniel Grant, the **Communications Manager**, issued client updates regarding service continuity.
 - Fiona Yang, the **Public Relations Officer**, managed media queries with a prepared statement.
- **Outcome:**
 - Transparent communication minimized client concerns and preserved Liangshan's reputation.

13.6 Recovery and Post-Incident Actions

13.6.1 Operational Resumption

- **Timeline:**
 - By 9:00 AM, critical business operations were resumed from the standby facility.
- **Security Measures:**
 - Additional security controls, including enhanced monitoring and restricted access, were implemented at the standby facility.

13.6.2 Site Rebuilding

- **Actions:**
 - The Facilities Team, in coordination with **CarePoint Maintenance**, began repair work at the primary data center.
 - Structural enhancements, such as improved fire detection and suppression systems, were planned.

14. Post-Incident Review

- **Comprehensive Analysis:**
 - The Disaster Recovery Lead (Hannah Li) conducted a post-mortem review.
 - The review identified gaps in fire detection and suppression systems at the primary facility.
- **Policy Updates:**
 - Fire suppression protocols and DRP communication plans were revised.
- **Training:**
 - Refresher training sessions were scheduled for all DR teams.

15. Conclusion

Liangshan Enterprise's Disaster Recovery Plan successfully minimized the operational impact of the fire outbreak. The activation of the standby facility and coordinated efforts of all DR teams ensured critical business continuity within hours, reaffirming Liangshan's commitment to resilience and client trust.